Linear Regression Analysis in the Social Sciences

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Selected Topics

Goals

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1. How do we use OLS with panel data?

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2. What are applications of Two-Stage models?

Panel Data Regression Models

- Time series data: values of one or more variables over time
- Cross-section data: values of one or more variables for several sample units at the same point in time
- Panel data: same cross-sectional unit surveyed over time
- Advantages of panel data:
  - Account for heterogeneity among units
  - More informative data, more variability, less collinearity, more degrees of freedom, more efficiency
  - Better suited to study dynamics of change
  - Can detect and measure effects not observable in pure cross-section or time series data
  - Enables studying more complicated behavioral models
  - Minimizes bias from aggregating individuals or firms into broad aggregates

Pooled OLS Regression

- Consider the model:
$$C_{it} = \beta_1 + \beta_2 Q_{it} + \beta_3 PF_{it} + \beta_4 LF_{it} + u_{it}$$
- Pooled OLS regression assumes the same regression coefficients for all airlines
- Potential problem: does not distinguish between airlines or account for heterogeneity
- Error term may be correlated with regressors, leading to biased and inconsistent estimates

```
# Load the plm package
library(plm)

# Assuming the data is a data frame called "airlines"
# Perform pooled OLS regression
pooled_model <- plm(c ~ q + pf + lf, data = airlines
, model = "pooling")
summary(pooled_model)
```

The Fixed-Effect Models

- Fixed-effect model (least-squares dummy variable model) allows for heterogeneity among subjects
- Each entity has its own intercept value
- Model:

$$C_{it} = \beta_{1i} + \beta_2 Q_{it} + \beta_3 PF_{it} + \beta_4 LF_{it} + u_{it}$$

- Intercept α_{ji} is time-invariant for each airline
- Estimate using dummy variables:

$$\begin{aligned} C_{it} = \alpha_1 + \alpha_2 D_{2i} + \alpha_3 D_{3i} + \alpha_4 D_{4i} + \alpha_5 D_{5i} + \alpha_6 D_{6i} \\ + \beta_2 Q_{it} + \beta_3 PF_{it} + \beta_4 LF_{it} + u_{it} \end{aligned}$$

```
# Perform fixed-effect regression
fixed_model <- plm(c ~ q + pf + lf, data = airlines
, model = "within")
summary(fixed_model)
```

A Caution in the Use of the Fixed Effect LSDV Model

- Problems with fixed effects models:
  - Too many effects may lead to degrees of freedom problem
  - Possibility of multicollinearity with many dummy variables
  - May not be able to identify the impact of time-invariant variables
- Example: War = f(democracy)
  - FE model will drop war and democracy observations for units that don't vary

The Random Effects Model (REM)

- Express lack of knowledge about the true model through the disturbance term
- Model:

$$C_{it} = \beta_1 + \beta_2 Q_{it} + \beta_3 PF_{it} + \beta_4 LF_{it} + \epsilon_i + u_{it}$$

- ϵ_i is a random error term with mean zero and variance σ^2_ϵ
- Composite error term $w_{it} = \epsilon_i + u_{it}$
- Assumptions:
  - $\epsilon_i \sim N(0, \sigma^2_\epsilon)$
  - $u_{it} \sim N(0, \sigma^2_u)$
  - $E(\epsilon_i u_{it}) = 0$
  - $E(\epsilon_i \epsilon_j) = 0$, for $i \neq j$
  - $E(u_{it} u_{is}) = E(u_{jt} u_{js}) = E(u_{it} u_{js}) = 0$, for $i \neq j$ and $t \neq s$

```
# Perform random-effect regression
random_model <- plm(c ~ q + pf + lf, data = airlines
, model = "random")
summary(random_model)
```

The Hausman Test: The Fixed-Effect Models vs. The Random Effect Models

- Hausman test: choose between fixed-effect and random-effect models
- Null hypothesis: FEM and REM estimators do not differ substantially
- If null hypothesis is rejected, REM is not appropriate (random effects correlated with regressors)
- Fixed-effect model is preferred in this case

```
# Perform Hausman test
phtest(fixed_model, random_model)
```

The Breusch-Pagan Test: Testing for Random Effects

- Breusch-Pagan (BP) test: choose between random-effect model and simple OLS model
- Null hypothesis: variances across entities is zero (no panel effect)
- If null hypothesis is rejected, use panel estimations (fixed-effect or random-effect models)
- If null hypothesis is not rejected, random effects is not appropriate (use simple OLS regression)

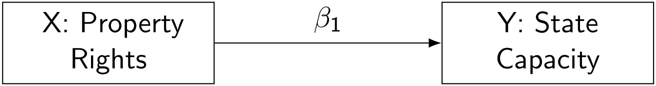
```
# Perform Breusch-Pagan test
plmtest(random_model, type =
 "bp")
```

Two-stage Models

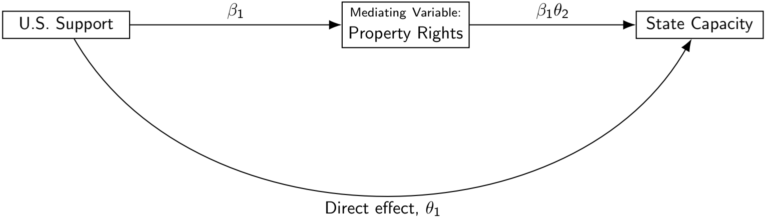
Goals

1. Understand the logic of multi-stage models
2. Examine applications of two-stage models:
    1. Mediation models
    2. Selection models
    3. Instrumental variable models

Models simplify the world

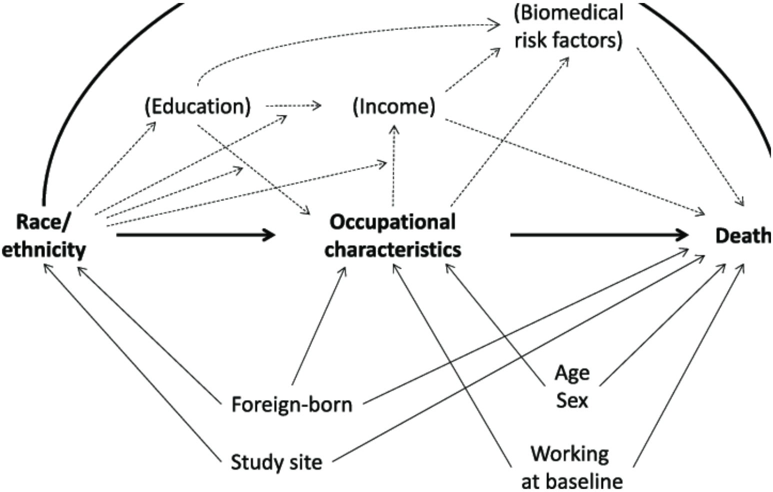


But models shouldn't be too simple



Modeling the world

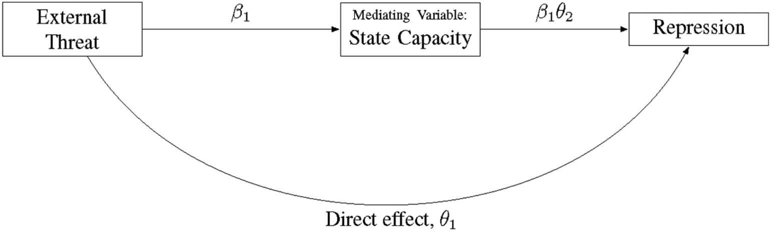
1. Given that the world is complicated, we should use various tools to help us model.
2. One important tool is a Directed Acyclic Graph, or simply DAG.
3. This can help us think about the relationships between variables.



(Biomedical
risk factors)
(Education)
(Income)
Race/
ethnicity
Occupational
characteristics
Death
Foreign-born
Study site
Age
Sex
Working
at baseline

How DAGs Help

1. Help identify potential confounders (remember OLS assumptions!!)
2. Help think about direct and indirect effects.



95% confidence intervals

Mediation Analysis

1. The aim is to understand if and to which extent the effect of a treatment variable **T** on an outcome variable **Y** is mediated through a variable **M**
2. Requires 2 (or more) equations.
    1. An equation for the outcome **Y**, as a function of M and T
    2. An equation for the Mediator **M**, as a function of T

Mediation Analysis Example

$$
\begin{aligned}
E[\text{Repression} \mid \text{External Threat}, \text{Capacity}, C] = & \theta_0 + \theta_1 \text{External Threat} \\
& + \theta_2 \text{Capacity} \\
& + \theta_j C + \mu
\end{aligned} \tag{1}
$$

$$
E[\text{Capacity} \mid \text{External Threat}, C] = \beta_0 + \beta_1 \text{External Threat} + \beta_j C + \upsilon \tag{2}
$$

Mediation Analysis

1. We are interested in 3 estimates (but only need to calculate 2).
    1. Total effect of T (i.e. Direct + Indirect Effect)
    2. Direct effect of T
    3. Indirect Effect of T, mediated by M

Mediation Analysis: Total Effect

1. Total effect of T can be estimated with θ_1 in the following equation (excluding the mediator):

$$E[\text{Repression} \mid \text{External Threat}, C] = \theta_0 + \theta_1 \text{External Threat} + \theta_j C + \mu \tag{3}$$

Mediation Analysis: Direct Effect

1. Direct effect of T can be estimated with θ_1 in the following equation (including the mediator):

$$\begin{aligned} E[\text{Repression} \mid \text{External Threat}, \text{Capacity}, C] = & \theta_0 + \theta_1 \text{External Threat} \\ & + \theta_2 \text{Capacity} \\ & + \theta_j C + \mu \end{aligned} \tag{4}$$

Mediation Analysis: Indirect Effect

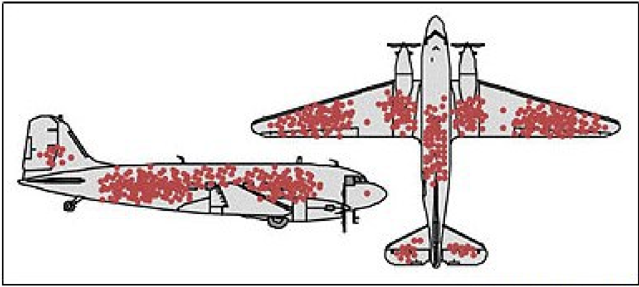
1. Indirect effect of T can be estimated with $\theta_2 \times \beta_1$ in the following equations (including the mediator):

$$\begin{aligned} E[\text{Repression} \mid \text{External Threat}, \text{Capacity}, C] =& \theta_0 + \theta_1 \text{External Threat} \\ & + \theta_2 \text{Capacity} \\ & + \theta_j C + \mu \end{aligned} \tag{5}$$

$$E[\text{Capacity} \mid \text{External Threat}, C] = \beta_0 + \beta_1 \text{External Threat} + \beta_j C + \upsilon \tag{6}$$

Mediation Analysis in R

1. mediate



Credit: Cameron Moll

Selection Models

1. Suppose we're interested in explaining why wars end (or escalate) as a function of T
2. Empirically, we can only observe T during wars but not in wars that never started.
3. This creates bias if T explains why wars begin in the first place.

Selection Models

1. To address potential selection bias, we can model whether an observation ends up in the sample (i.e. war) to begin with.
2. Then we can model an outcome, conditional on whether the observation is likely to be in the sample or not.
3. Many ways to model this selection process, but the Heckman model is canonical.

   Heckman, James. 1979. "Sample Selection Bias as a Specification Error" Econometrica, 47(1): 153-162.

Heckman Selection Model

Outcome stage:

$$Y = \beta_1 X_1 + IMR + \mu_1 \tag{7}$$

Selection stage:

$$S = \theta_1 X_1 + \theta_2 Z_1 + \mu_2 \tag{8}$$

1. where **IMR** is the **inverse Mill's Ratio** derived from the selection stage.
2. The selection stage must include one variable that is not included in the outcome stage and is unrelated to the outcome (Z).
3. μ_1 and μ_2 must be unrelated to each other.
4. Don't calculate manually! SE's will be wrong.

Selection Models in R

1. sampleSelection, selection
2. vglm, tobit (for censored data)
3. heckit (for binary outcomes)

Instrumental Variable Models

- Fundamental Problem of Causal Inference

$$Y = \alpha + \beta_1 X + \varepsilon,$$

- If X is binary, We can then estimate the average treatment effect as

$$\beta_1 = E[Y \mid X = 1] - E[Y \mid X = 0]$$

- But we cannot observe $X = 1$ and $X = 0$ for the same observation.
- In other words, we cannot observe the counterfactual (i.e. we have a missing data problem).

Fundamental Problem of Causal Inference

Example

- Suppose after administering a test to a group of young adults, we find that individuals who have attended college score higher than individuals who have not attended college.
- What explained these observed effects?

EXAMPLE	
Individuals that attend college show higher mental ability	Observed Differences
Attending college makes individuals smarter	Average Treatment Effects
Individuals who attend college are smarter in the first place	Initial differences/Selection
The mental ability of those who attend college may increase more than would the mental ability of those who did not attend college if they had instead attended college.	Differential Treatment Effects

Fundamental Problem of Causal Inference

- Experiments are the gold standard of addressing these problems
- Research controls the randomization of the treatment
- By definition, the treatment assignment is unrelated to the error term.

Instrumental Variable Models

- We don't usually have the luxury to randomize something like college attendance.
- Other social science behavior cannot be studied in an experiment format.
- Instrumental Variable Models (IV) are one way to address this problem.

Instrumental Variable Models

The basic idea is that if we can replace the actual realized values of x_i (which are correlated with u_i) by predicted values of x_i that are

- related to the actual x_i
- but uncorrelated with u_i
- The general problem in practice is finding instrumental variables that have both these properties (and note that the second property is highly theoretical, and thus hard to defend).

Instrumental Variable Models

$$x = \theta_1 z_1 + \nu$$

If z is a valid instrument, it gives us an exogenous prediction of x (x^*), which we assume is "as-if" random.

$$y = \beta_1 x_1^* + \mu$$

Instrumental Variable Models in R

Commands

- ivreg
- fixest, feols